Lecture Scribe: Joint Random Variables + Joint Distributions + Examples

Course: CSE400 – Fundamentals of Probability in Computing **Instructor:** Dhaval Patel, PhD
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1. Introduction to Joint Random Variables

- **Motivation:** In science and real life, we are often interested in two or more random variables at the same time.
- **Purpose:** The random variables have a joint distribution that allows us to compute probabilities of events involving both variables and understand the relationship between them.
- **Dependence:** This is simplest when the variables are independent. When they are not, covariance and correlation are used as measures of the nature of the dependence between them.
- **Examples:**
  - Height and weight of giraffes.
  - Frequency of exercise and rate of heart disease.
  - Smart Transportation: X = Vehicle Speed and Y = Traffic Density.
  - Wireless Network Performance: X = Signal-to-Noise Ratio (SNR) and Y = Data Throughput. Throughput depends on SNR, making X and Y correlated random variables.
  - Weather Prediction: X = Rainfall and Y = Temperature.

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2. Discrete Case: Joint Probability Mass Function (PMF)

- **Setup:** Suppose X and Y are discrete random variables where $X \in \{x_1, x_2, \ldots, x_n\}$ and $Y \in \{y_1, y_2, \ldots, y_m\}$.
- **Product Set:** The ordered pair (X, Y) takes values in the product set:
  $\{(x_1, y_1), (x_1, y_2), \ldots, (x_n, y_m)\}$.
- **Definition (Joint PMF):** $p(x_i, y_j) = P(X = x_i, Y = y_j)$. This gives the probability of the joint outcome occurring simultaneously.
- **Properties:**
  - $0 \leq p(x_i, y_j) \leq 1$.
  - $\sum_{i=1}^{n} \sum_{j=1}^{m} p(x_i, y_j) = 1$.
- **Example (Rolling Two Dice):** * **Experiment:** Roll two fair dice. Let X = value on the first die, and T = total (sum) of both dice.
  - Since each outcome (i, j) has probability $\frac{1}{36}$, we can construct the joint probability table.
  - **Observations:** Entries in the table are either 0 or $\frac{1}{36}$, and X and T are not independent.

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3. Continuous Case: Joint Probability Density Function (PDF)

- **Analogy:** The continuous case is analogous to the discrete case, where discrete values become continuous intervals, Joint PMF becomes Joint PDF, and sums become double integrals.
- **Setup:** If $X \in [a, b]$ and $Y \in [c, d]$, then the pair (X, Y) takes values in the product region: $[a, b] \times [c, d]$.
- **Definition (Joint PDF):** A function $f(x, y)$ is called the joint probability density function of X and Y if $P((X, Y)$ lies in a small rectangle of area $dx\,dy) \approx f(x, y)dx\,dy$.
- **Key Properties:**
  - $f(x, y) \geq 0$.
  - $\int_c^d \int_a^b f(x, y)\,dx\,dy = 1$.
  - The joint PDF $f(x, y)$ may take values greater than 1. It is a probability density, not a probability.

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4. Worked Example: Continuous Joint PDF

- **Problem:** Given $f(x, y) = 4xy$, where $0 \leq x \leq 1$ and $0 \leq y \leq 1$. Jointly, (X, Y) takes values in the unit square $[0, 1]^2$.
- **Task 1: Show that $f(x, y)$ is a valid joint PDF.**
  - **Step 1:** Check Validity (Normalization).
  - $\int_0^1 \int_0^1 4xy\,dx\,dy = \int_0^1 \left(\int_0^1 4xy\,dx\right) dy = \int_0^1 2y\,dy = \left[y^2\right]_0^1 = 1$.
  - **Conclusion:** Hence $f(x, y) \geq 0$ and total probability is 1. So it is a valid joint PDF.
- **Task 2: Compute $P(A)$ for the event $A = \{X < 0.5, Y > 0.5\}$.**
  - **Step 2:** Compute $P(A)$.
  - $P(A) = \int_0^{0.5} \int_{0.5}^1 4xy\,dy\,dx = \int_0^{0.5} \left[2xy^2\right]_{0.5}^1 dx = \int_0^{0.5} 2x(1 - 0.25)\,dx = \int_0^{0.5} \frac{3}{2}x\,dx = \left[\frac{3}{16}\right]$.

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5. Joint Cumulative Distribution Function (Joint CDF)

- **Notation:** We use the notation $\{X \leq x$ and $Y \leq y\}$ to mean the event $X \leq x, Y \leq y$.
- **Definition:** The joint cumulative distribution function is defined as $F(x, y) = P(X \leq x, Y \leq y)$.
- **Continuous Case:** * If X and Y are continuous random variables with joint density $f(x, y)$ over $[a, b] \times [c, d]$, the joint CDF is obtained by integrating the density.
  - $F(x, y) = \int_c^y \int_a^x f(u, v)\,du\,dv$.
  - **Recovering the Joint PDF:** Since there are two variables, we use partial derivatives: $f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$.
- **Discrete Case:**
  - If X and Y are discrete random variables with joint PMF $p(x_i, y_j)$, the joint CDF is obtained by summing probabilities.
  - $F(x, y) = \sum_{x_i \leq x} \sum_{y_j \leq y} p(x_i, y_j)$.
  - **Interpretation:** Add all probabilities in the table for rows with $x_i \leq x$ and columns with $y_j \leq y$.

- **Properties of the Joint CDF:** The joint CDF $F(x, y)$ of X and Y must satisfy several properties:
  - $F(x, y)$ is non-decreasing. If x or y increase, then $F(x, y)$ must stay constant or increase.
  - $F(x, y) = 0$ at the lower-left of the joint range. If the lower-left corner is $(-\infty, -\infty)$, then $\lim_{(x,y)\to(-\infty,-\infty)} F(x, y) = 0$.
  - $F(x, y) = 1$ at the upper-right of the joint range. If the upper-right corner is (∞, ∞), then $\lim_{(x,y)\to(\infty,\infty)} F(x, y) = 1$.